Highlights

Reinforcement Learning vs. Rule-Based Policies for Dynamic Credit Threshold Control: A Simulation Study

- Weekly credit threshold control is cast as an MDP with delayed repayment outcomes.
- Six deep RL agents are benchmarked against five rule-based policies in six scenarios.
- Rule-based weekly search wins the reward ranking in every scenario and dimension.
- Training budget, not an RL ceiling, drove the profit gap: 9.4% falls to 0.5–2.5%.
- At 30D and 50D, RL cuts default rates below profit-seeking baselines in 5 of 6.